Rotaract Club of Xiamen University Malaysia

Rotary Club of Kuala Lumpur Dynamics

Rotary Club of Tun Razak Exchange

SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: [Insert date]

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick “No” or write “None” – do not leave it blank. Submit as a single PDF document named “SDC2026_TeamName_TechDeclaration.pdf” together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

Instagram: @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

1. TEAM AND PROJECT OVERVIEW

Team Name

Beginathon

Project / Solution Name

Just Right: AI bake planner for small cafés and bakeries

Name (Leader)

Derek Chay Wen Hong

Email & Phone (Leader)

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University Affiliation(s)

Multimedia University (MMU), Cyberjaya

Live Prototype URL

https://just-right-linegidid.replit.app · bot: t.me/justright_bakery_bot

Code Repository URL (public / access granted to judges)

https://github.com/derkglova/just-right

2. ARCHITECTURE & TECHNOLOGY STACK

2.1. System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

Staff, manager and baker use a Telegram bot (Telegram Bot API). At closing, the closing count goes to the backend, a Python/Flask app that stores daily sales, plans and an activity log in SQLite. The AI forecast (scikit-learn, one linear model per item) retrains on that history together with weather from the Open-Meteo API and predicts tomorrow's demand. A newsvendor rule turns the prediction into a bake quantity. The bot sends the plan to the manager, who approves or edits it; the baker then receives the list at a set time (APScheduler).

A manager dashboard (HTML, React and Chart.js served by Flask) reads the same data through a JSON API. Everything runs as one process on a Replit Reserved VM deployment; secrets live in Replit Secrets. Orbyte data (items, costs, roles, POS sales) is simulated. A block diagram is attached as Appendix A (last page).

2. ARCHITECTURE & TECHNOLOGY STACK

2.2. Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
Python 3.12	Main language (backend, bot, AI)	PSF License
Flask 3.0.3; APScheduler 3.10.4	Web server, JSON API, dashboard; daily timers	BSD-3-Clause; MIT
pyTelegramBotAPI 4.21.0	Telegram bot client	GPL-2.0
scikit-learn 1.5+; NumPy	AI demand forecast model	BSD-3-Clause
requests 2.31+; python-dotenv 1.0.1	Calls the Open-Meteo API; local settings	Apache-2.0; BSD-3-Clause
React; Chart.js 4.4.1 (CDN); SQLite	Dashboard UI and charts; database	MIT; public domain

3. APIs & CLOUD SERVICES

3.1. External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API and Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
Telegram Bot API (Telegram)	Messages, buttons and commands for staff, manager and baker	Free
Open-Meteo Forecast API (Open-Meteo)	Past weather for training and tomorrow's forecast for Cyberjaya	Free (no key)
No GenAI / LLM APIs	The AI model runs inside our own app	None

3. APIs & CLOUD SERVICES

3.2. Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit Deployments (Replit)	Hosts bot, dashboard, API and timers as one always-on app; deps installed with uv sync	Reserved VM, 0.5 vCPU / 2 GiB, Asia
Replit Secrets (Replit)	Stores Telegram token and user IDs	Included with Replit Core
SQLite on the deployment	Database (sales, plans, activity log)	Same Replit VM
GitHub (GitHub)	Source code and version control; pulled into Replit to deploy	Free

4. AI TOOLS, MODELS & DATASETS

4.1. AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
Linear regression demand model (one per item)	scikit-learn; trained by our team on our data	Self-trained; retrained after every closing count	BSD-3-Clause (library)

4. AI TOOLS, MODELS & DATASETS

4.2. Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Synthetic sales history (120 days, 3 items)	scripts/generate_data.py in our repo	Synthetic	Own work (Apache-2.0)	N
Weather history + forecast, Cyberjaya	open-meteo.com	Public	CC BY 4.0	N
Items, costs, staff roles (simulated Orbyte)	src/backend/data in our repo	Synthetic	Own work (Apache-2.0)	N

4. AI TOOLS, MODELS & DATASETS

4.3. AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☒ Yes ☐ No

If yes, list them below:

Tool & Version	Used For	Extent of Use (minor / moderate / major)
Claude Code (Anthropic)	Writing and debugging backend, bot and dashboard code	Major
Claude (Anthropic) chat	Brainstorming, critique, planning, documentation drafts, starter planner and forecast code	Major
Claude design (Anthropic)	Dashboard visual design mock-up	Moderate
Replit Agent	Initial project import set-up (its changes were later discarded)	Minor

4. AI TOOLS, MODELS & DATASETS

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

The problem, solution design and technical decisions (newsvendor rule, forecast features, roles, scope) were made by our team. We read and edited all AI-generated code, ran unit tests for the planner, forecast and database, back-tested the model on historical data, and tested the bot and dashboard end to end on Replit before submitting. Each member can explain every major part.

5. SPONSOR-PROVIDED RESOURCES

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☒ Yes ☐ No

If yes, list them below:

Sponsor	Resource (API / credits / platform / data)	How It Was Used in the Project
Replit	Replit Core (platform + credits)	Hosts the deployed app (Reserved VM), Secrets, Git sync. Proof: submission/SDC 2026_Beginathon_Replit_POU.pdf
SDC mentors (KrackedDevs)	Mentoring session	15-min idea review with Muhammad Fadli bin Md Khalid, 21 Oct 2026
RunPod	GPU credit	Not used: no GPU workload

5. SPONSOR-PROVIDED RESOURCES

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Replit is essential to how the prototype is delivered: it hosts the always-on bot, dashboard and timers. Technically it could be replaced by another always-on Python host, but Replit gave us one-click deployment, secrets management and Git sync.

6. SECURITY, PRIVACY & RESPONSIBLE USE

6.1. Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

No customer data is collected. The app stores shop data only: daily bake, sales and leftover counts, plans, and an activity log. For staff it stores their Telegram user ID, display name and role, needed to control who can approve plans.

Data lives in SQLite on the Replit deployment and is kept for the duration of the prototype. The Telegram token and user IDs are kept in Replit Secrets, never in the code. No data is sent to AI APIs: the model runs inside our app. Open-Meteo only receives the shop's coordinates.

In a real deployment we would follow Malaysia's PDPA principles: collect only what is needed, tell staff what is stored, and let the owner delete records.

6. SECURITY, PRIVACY & RESPONSIBLE USE

6.2. Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

The demo uses synthetic sales data, so the accuracy figures (2.4 vs 3.2 pieces) show the method, not real-shop performance. A linear model is simple and explainable but can miss unusual patterns; it needs about 14 days of history before it beats a simple average, so new shops start in a learning mode.

Weather forecasts can be wrong; changes are capped at ±15% per day and the manager approves every plan. Sell-out times are estimated from rough time buckets. If Open-Meteo is down, cached weather is used. Only one bot instance may poll Telegram at a time, so we run a single always-on deployment.

7. COMPLIANCE & READINESS

Tick all boxes that apply before submitting.

☒ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.

☒ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.

☒ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.

☒ Any use of generative-AI tools for development has been disclosed and reviewed by the team.

☒ Any use of official SDC sponsor resources has been declared in Section 5.

☒ The prototype described here is accessible to judges through the links provided in Section 1.

☒ All fields in this template have been completed within the word limits stated.

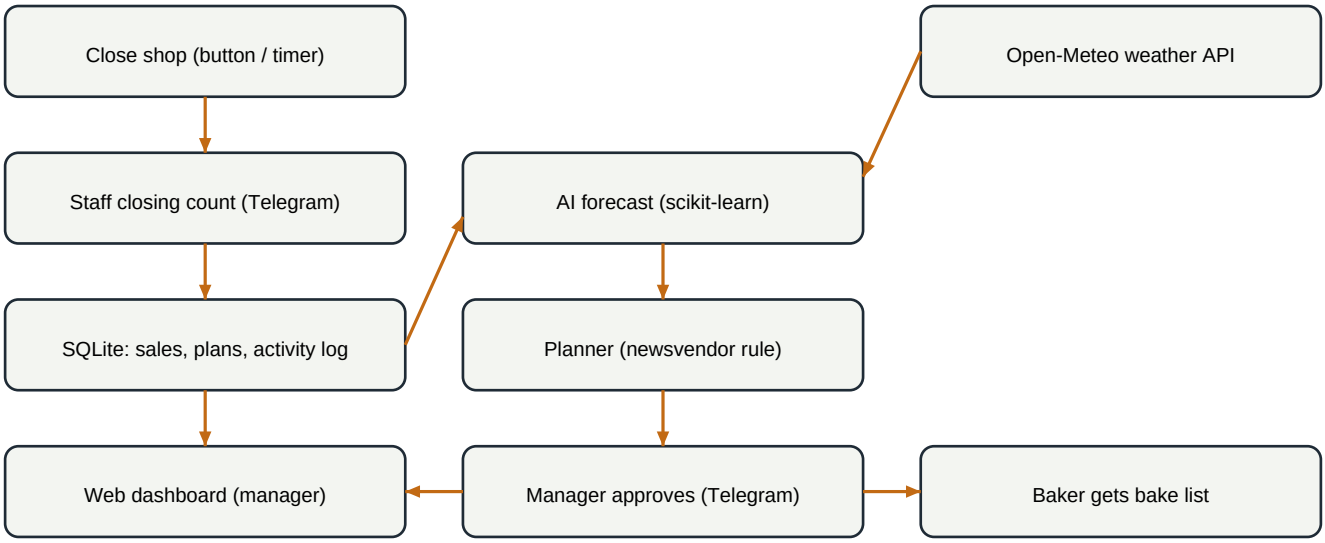
8. DECLARATION

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

[Signature]
Derek Chay Wen Hong, Team Leader

[Date]
24 October 2026

Appendix A: System Block Diagram



Everything runs as one Python process on Replit (Reserved VM). Orbyte data (sales, costs, roles) is simulated.